

Supporting information for:

Theory for the Conformation of Adhesion
Protein

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Free Energy of Adhesion Proteins

The free energy of the single protein before binding includes the conformation entropy and translational entropy. Due to the fact that one end is adhere to the membrane, the translational entropy is

$$-TS_t = k_B T \ln \sigma b^2 - 1 \quad (\text{S1})$$

where $\sigma = n/A$ is the graft density, while b^2 is the area of a single protein for the dimensionless. For the contribution of conformation energy, the chain is in the free state because of one free end of the protein. The free energy is about $k_B T$. The free energy of the single protein before binding is thus

$$F_1 = F_t + F_c = k_B T \ln \sigma b^2 \quad (\text{S2})$$

After binding with the other protein, two ends are adhered to the membranes, where the protein is likely stretched by the two membranes. The translational entropy is the same as that before binding with the expression of

$$F_t = -TS_t = k_B T \ln \sigma b^2 - 1 \quad (\text{S3})$$

Using the blob scaling, the free energy of a stretching protein is

$$F_c = k_B T \left(\frac{D}{R_g} \right)^2 \quad (\text{S4})$$

where D is the membrane distance while R_e is the end-to-end distance of a single protein in the free state with the express of

$$R_g^2 = L^2 f(x) = 2l_p^2 (e^{-x} - 1 + x) \quad (\text{S5})$$

$f(x)$ is the Debye function, l_p is the persistent length of the protein and $x = L/l_p$. The free

energy after binding is thus

$$F_2 = F_t + F_c = k_B T \ln \sigma b^2 - 1 + k_B T \left(\frac{D}{R_g} \right)^2 \quad (S6)$$

The formation of binding proteins is thus

$$\Delta F = F_2 - 2F_1 = k_B T \left(\frac{D}{R_e} \right)^2 - k_B T \ln \sigma b^2 - 1 - F_{bind} \quad (S7)$$

Then we discuss three typical cases : (a) rigid ($l_p = 100nm$), (b) semi-flexible ($l_p = 10nm$), and (c) flexible proteins ($l_p = 1nm$). The R_e is $14.63nm$, $12.02nm$, and $5.29nm$. The membrane distances are $15.5nm$, $12.5nm$, and $8.5nm$. The free energy due to the conformation change is $F_a = (15.5/14.63)^2 = 1.12$, $F_b = (12.5/12.02)^2 = 1.08$, and $F_c = (8.5/5.29)^2 = 2.58$.

Then we discuss three typical cases : (a) rigid ($l_p = 86.75nm$), (b) semi-flexible ($l_p = 8.37nm$), and (c) flexible proteins ($l_p = 1.15nm$). The R_e is $14.58nm$, $11.59nm$, and $5.64nm$. The membrane distances are $15.5nm$, $12.5nm$, and $8.5nm$. The free energy due to the conformation change is $F_a = 1.5 \times (15.5/14.58)^2 = 1.69$, $F_b = 1.5 \times (12.5/11.59)^2 = 1.74$, and $F_c = 1.5 \times (8.5/5.64)^2 = 3.4$. So $F_c - F_a = 1.70$, $F_c - F_b = 1.65$, and $F_b - F_a = 0.05$

$K_{2Da} = 12613$, $K_{2Db} = 876$, and $K_{2Dc} = 359$. We have $\ln(K_{2Da}/K_{2Db}) = F_b - F_a = 2.66$, $\ln(K_{2Db}/K_{2Dc}) = F_c - F_b = 0.89$, $\ln(K_{2Da}/K_{2Dc}) = F_c - F_a = 3.56$,

For the rigid case: there are 7 bonds with length about $1nm$, the total length is about $L = 15nm$, the persistent length chose is $100nm$, gives $L/l_p = 0.15$ and $R_e = 14.63nm$.

$$\frac{K_a}{K_b} = \exp[-(F_a - F_b)/k_B T] \quad (S8)$$

Using the force vs extension

$$\frac{fb}{k_B T} = \frac{3 \langle R \rangle}{R_{max}} \quad (S9)$$

The free energy is

$$F = \frac{f}{k_B T} \langle R \rangle = \frac{3 \langle R \rangle^2}{R_{max} b} \quad (S10)$$

where $b = 2l_p$ for the worm-like chain model. For the strong stretching,

$$\frac{fb}{k_B T} = \frac{1}{2} \left(\frac{R_{max}}{R_{max} - \langle R \rangle} \right)^2 \quad (\text{S11})$$

The free energy is

$$F = \frac{f}{k_B T} \langle R \rangle = \frac{\langle R \rangle}{2b} \left(\frac{R_{max}}{R_{max} - \langle R \rangle} \right)^2 \quad (\text{S12})$$

Bonding Probability

Considering the effect of chain stiffness on the bonding probability. 1. Rigid Polymer: the bonding end moves in the mid-planar of the membrane, the probability is

Just as hydrogen bonds are formed between donors and acceptors, so here we can term the up ions as donors and the bottom segments as acceptors. We start with the n_0 unbound lithium ions. The number of ways to choose n_b bonds is

$$W_1 = \frac{n_0!}{n_b!(n_0 - n_b)!} \quad (\text{S13})$$

The number of ways of linking the donors with acceptors is

$$W_2 = n_b! \quad (\text{S14})$$

The formation of a bond requires the donor and receptor pair to be in close proximity within some volume scale v_b . For simplicity, we take v_b to be just the monomer volume v (I think the v_b is different for the semi-flexible, and rigid polymer).

$$P = \left(\frac{v}{V}\right)^{n_b} \quad (\text{S15})$$

The overall probability of forming n_b bonds is then

$$f^*V = -\ln(W_1 W_2 P) \quad (\text{S16})$$

Assuming all the chains are bonded, the free energy is

$$f^*V = -n_b \ln \frac{n_b v}{V} + n_b \quad (\text{S17})$$

For the rigid polymer, the bond only happens on a 2D mid-planar of the membrane, thus

$$\frac{n_b v}{V} = \frac{n_b b^2}{A} \quad (\text{S18})$$

For the semi-flexible and flexible protein, the binding happens at a 3D space because the end can be fluctuated in the z-axis. Thus the probability is decreased with the formula as

$$\frac{n_b v}{V} = \frac{n_b b^3}{AL} \quad (\text{S19})$$

where $L = R_{max} - L_m$ (L_m is the membrane distance and R_{max} is the maximum length of the protein) is distance that the chain end can distribute in the z-axis due to the flexibility of the protein. $b \approx 0.5nm$ is the diameter of the bead.

for the semi-flexible and flexible chain, $L = 16 - 13.23 = 2.77nm$ and $L = 16 - 8.22 = 7.78nm$ respectively. Thus the

$$F_b - F_a = 1.71 \quad (\text{S20})$$

and

$$F_c - F_b = 1.0 \quad (\text{S21})$$

Considering that the bind probability decreases as chain becomes flexible, thus $F_b - F_a$ is larger than 1.71.

Plus the loss of rotational entropy after binding, for the rigid polymer

$$-S_{rot}^{rigid} = -\ln \frac{\omega_b \omega_{RL}}{\omega_L \omega_R} = -\ln \frac{0.33\omega_b}{0.75 \times 0.75} = -\ln 0.586 \times \omega_b \quad (\text{S22})$$

for the semi-flexible polymer, the rotational entropy is

$$-S_{rot}^{semi-rigid} = -\ln \frac{\omega_b \omega_{RL}}{\omega_L \omega_R} \quad (\text{S23})$$

$\frac{\omega_{RL}}{\omega_L \omega_R} \approx 0.227$ for the seim-rigid

Invoking the identity using the average density field ρ , $F[\hat{\rho}] = \int D\rho F[\rho] \delta(\rho - \hat{\rho})$ and decoupling the two-body χ term by the *Hubbard–Stratonovich* transformation, the canonical partition function can be written

$$Z = \int D\rho \int D\omega \int D\eta \exp[-H(\rho, \omega, \eta)] \quad (\text{S24})$$

with an effective Hamiltonian given by

$$\begin{aligned} H(\rho, \omega, \eta) = & \int d\mathbf{r} \left[\chi_{AB} v_0 (\rho_{1A}(\mathbf{r}) + \rho_{2A}(\mathbf{r})) (\rho_{1B}(\mathbf{r}) + \rho_{2B}(\mathbf{r})) \right. \\ & + \chi_{As} v_0 (\rho_{1A}(\mathbf{r}) + \rho_{2A}(\mathbf{r})) \rho_s(\mathbf{r}) + \chi_{Bs} v_0 (\rho_{1B}(\mathbf{r}) + \rho_{2B}(\mathbf{r})) \rho_s(\mathbf{r}) \\ & - \omega_{1A} \rho_{1A}(\mathbf{r}) - \omega_{1B} \rho_{1B}(\mathbf{r}) - \omega_{2A} \rho_{2A}(\mathbf{r}) - \omega_{2B} \rho_{2B}(\mathbf{r}) - \omega_s \rho_s(\mathbf{r}) \\ & + \eta(\mathbf{r}) (\rho_{1A}(\mathbf{r}) + \rho_{2A}(\mathbf{r}) + \rho_{1B}(\mathbf{r}) + \rho_{2B}(\mathbf{r}) + \rho_s(\mathbf{r}) - 1/v_0) \\ & \left. - \frac{F_1(\mathbf{r})}{n_1 f_B N_1} (\mathbf{r} \rho_{1B}(\mathbf{r}) - \rho_{01B} \xi_1) - \frac{F_2(\mathbf{r})}{n_2 f_B N_2} (\mathbf{r} \rho_{2B}(\mathbf{r}) - \rho_{02B} \xi_2) \right] \\ & - \ln Q_1 - \ln Q_2 - n_s \ln Q_s \end{aligned} \quad (\text{S25})$$

The density fields ρ_{1A} , ρ_{2A} , ρ_{1B} , ρ_{2B} and ρ_s are the ensemble averages of the microscopic density fields subject to the fields ω_{1A} , ω_{1B} , ω_{2A} , ω_{2B} , and ω_s . The field $\eta(\mathbf{r})$ is a Lagrange multiplier used to enforce the incompressibility constraint. We have absorbed the imaginary i into the fields ω_1 , ω_2 , ω_s , and $\eta(\mathbf{r})$ in anticipation that the saddle point values of the original fields lie on the imaginary axis.

where Q_1 , Q_2 , and Q_s are the single-chain partition functions for the test chain and the solvent chain, respectively, with the expression

$$\begin{aligned} Q_1 &= \frac{1}{V} \int d\mathbf{r} q_1(\mathbf{r}, N_1) \\ Q_2 &= \frac{1}{V} \int d\mathbf{r} q_2(\mathbf{r}, N_2) \\ Q_s &= \frac{1}{V} \int d\mathbf{r} \exp[-\omega_s(\mathbf{r})] \end{aligned} \quad (\text{S26})$$

Note that both $\mathbf{F}_2$ and $\mathbf{r}$ are vectors, because the radical value of $\mathbf{F}_2$ are 0, the $\mathbf{F}_2\mathbf{r}$ are $F_z \times lz$

Coupling the effect of fixing center-of-mass, the modified diffusion functions for two chains are

$$\frac{\partial}{\partial s} q_1(\mathbf{r}, s; [\omega]) = \frac{b^2}{6} \nabla^2 q_1(\mathbf{r}, s; [\omega]) - \left(\omega(\mathbf{r}) - \frac{\mathbf{F}_1}{N_1}(\mathbf{r} - \xi_1) \right) q_1(\mathbf{r}, s; [\omega]), \quad (\text{S27})$$

and

$$\frac{\partial}{\partial s} q_2(\mathbf{r}, s; [\omega]) = \frac{b^2}{6} \nabla^2 q_2(\mathbf{r}, s; [\omega]) - \left(\omega(\mathbf{r}) - \frac{\mathbf{F}_2}{N_2}(\mathbf{r} - \xi_2) \right) q_2(\mathbf{r}, s; [\omega]), \quad (\text{S28})$$

respectively. we don't need to solve this modified diffusion equation

I think $\mathbf{F}_1$ and $\mathbf{F}_2$ both are vector and independent of r , however, the radical value of are 0. The mean-field expression for the partition function corresponds to the value of Eq. (S24) at the saddle point, $Z \approx \exp[-H(\rho^*, \omega^*, \eta^*)]$. The Helmholtz free energy is then

$$\begin{aligned} \beta F &= -\ln Z \approx H(\rho^*, \omega^*, \eta^*) \\ &= \int d\mathbf{r} [\chi v_0(\rho_1^* + \rho_2^*)\rho_s^* - \omega_1^*\rho_1^* - \omega_2^*\rho_2^* - \omega_s^*\rho_s^*] - \ln Q_1 - \ln Q_2 - n_s \ln Q_s \end{aligned} \quad (\text{S29})$$

For the numerical process, $\omega_1 + \omega_2 + 2\omega_s$, we obtain the

$$\eta(r) = \frac{1}{4}(\omega_1 + \omega_2 + 2\omega_s - 2\chi + \frac{\mathbf{F}_1\mathbf{r}}{n_1N_1} + \frac{\mathbf{F}_2\mathbf{r}}{n_2N_2}) \quad (\text{S30})$$

for the fixing center-of-mass condition, by multi $\int dr \rho_1(r)$, we obtain

$$\int dr \rho_1(r)(\omega_1^*(\mathbf{r}) = \chi v_0 \rho_s^*(\mathbf{r}) - \frac{\mathbf{F}_1\mathbf{r}}{n_1N_1} + \eta^*(\mathbf{r})) \quad (\text{S31})$$

Using the fixing center-of-mass condition, we have

$$\mathbf{F}_1 \xi_1 = \int dr \rho_1(r) (\chi v_0 \rho_s^*(\mathbf{r}) + \eta^*(\mathbf{r}) - \omega_1^*(\mathbf{r})) \quad (\text{S32})$$

and

$$\mathbf{F}_2 \xi_2 = \int dr \rho_2(r) (\chi v_0 \rho_s^*(\mathbf{r}) + \eta^*(\mathbf{r}) - \omega_2^*(\mathbf{r})) \quad (\text{S33})$$

or

$$\mathbf{F}_1 \xi_2 \frac{n_2 N_2}{n_1 N_1} = \int dr \rho_2(r) (\chi v_0 \rho_s^*(\mathbf{r}) + \eta^*(\mathbf{r}) - \omega_1^*(\mathbf{r})) \quad (\text{S34})$$

and

$$\mathbf{F}_2 \xi_1 \frac{n_1 N_1}{n_2 N_2} = \int dr \rho_1(r) (\chi v_0 \rho_s^*(\mathbf{r}) + \eta^*(\mathbf{r}) - \omega_2^*(\mathbf{r})) \quad (\text{S35})$$

These two equations seem unable to solve two Forces because $F_1 = F_2 = 0$ is one of the solution, another method to solve these equations are $F_1 = F_1 + wcmp \times k \int dr \rho_1^*(\mathbf{r})(r - \xi_2)$, where $k = 3k_B T / N_1 b^2$. This coefficient $wcmp$ should be very small.